

Viraj Sanghavi

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EDUCATION

Dwarkadas J. Sanghvi College of Engineering, Mumbai
Bachelor of Technology in Computer Engineering 2024 – Present
CGPA: 9.18

PROJECTS

ADAPT-Synthetix: Adaptive Closed-Loop ASR Framework | *Python, Wav2Vec2, FastAPI, PostgreSQL, Docker, React* 2025

- Built a research-grade speech pipeline linking Wav2Vec2 ASR with a multi-layer phoneme-level diagnostic engine and TTS-driven remediation loop.
- Implemented 8-feature acoustic noise fingerprinting (spectral centroid, MFCC variance, ZCR, harmonic ratio) to classify background conditions across 5 categories.
- Designed confidence-weighted priority queue that escalates low-confidence errors on safety-critical vocabulary above filler errors.
- Built phoneme-level drift monitoring with a 20-utterance rolling window; triggers LoRA fine-tuning data preparation when ≥ 5 phonemes enter high-risk state.
- Deployed full-stack system with FastAPI backend, React frontend, PostgreSQL persistence, Docker containerization, and CI/CD via GitHub Actions on Render.

Swipe to Export: Exporter-Importer Matching Model | *Python, Scikit-learn* 2024
LOC Hackathon

- Built ML model to match exporters with relevant importers from structured trade datasets under hackathon time constraints.
- Applied PCA for dimensionality reduction and implemented probabilistic scoring using Bayesian reasoning to rank match confidence.
- Focused on feature engineering, similarity-based ranking, and ranking optimization for real-world trade data.

Customer Churn Prediction | *Python, Scikit-learn* 2024

- Performed EDA to surface patterns driving customer churn in imbalanced dataset.
- Built logistic regression model with class weighting; evaluated using confusion matrix and recall-focused metrics.
- Analyzed false positive / false negative trade-off from a business cost perspective.

TECHNICAL SKILLS

Programming: Python, JavaScript

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Frameworks & Tools: FastAPI, React, Docker, Git, GitHub Actions

Databases: PostgreSQL, SQL

Concepts: Machine Learning, Feature Engineering, PCA, Bayesian Inference, CI/CD

EXTRACURRICULARS

DJS Anveshan | Research Club – Publicity & Research 2024 – Present

- Authored problem discovery report on a critical failure mode in Federated Learning for cross-institutional fraud detection specifically, robust aggregation filters silently discarding valid non-IID updates from minority institutions.
- Identified gap in existing FL security literature: no standardized pre-deployment verification protocol exists for this failure class as of 2025, citing IEEE Transactions on Big Data and JPMorgan's Project Aikya.